CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Physiotherapy (B.P.T)

Semester – V Examination – Nov 2011

PT301 ELECTROTHERAPY II

Date: 30.11.11, Wednesday **Time: 1.30p.m to1.50 p.m** **Maximum Marks: 10**

Instructions:

1. There are two parts in the paper. **PART –I** contains 10 Questions and **PART – II** contains 4 Questions.
2. **PART –I** should be answered in MCQ 'S question paper & **PART – II** should be answered in two separate answer books which has two sections.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **PART – I** is 20 min. Question paper for **PART – II** will be given at the table when the answer sheet for **PART – I** is returned. However, Candidates who complete **PART – I** earlier than 20 min, can collect main question paper for **PART –II** immediately after handing over the **PART-I** question paper.

PART-I

Multiple Choice Questions: (MCQs) **(1 × 10 = 10)**
(Encircle the correct answer)

1. The resting membrane potential has.......
   a. Similar charges intra & extracellularly.
   b. Positivity outside & negativity inside.
   c. Negativity outside & positivity inside.
   d. None of the above.

2. The term Orthodromic implies......
   a. Passage of impulse in forward direction.
   b. Passage of impulse in the opposite direction.
   c. Passage of impulse in both the direction.
   d. None of the above.

3. The resistance offered by the epidermis to the flow of electric current is ....
   a. 10 Ohms.
   b. 100 Ohms.
   c. 1000 Ohms.
   d. 100,00 Ohms.

4. Electrical stimulation of innervated muscles causes.......
   a. Initially activates the slow twitch muscle fibres.
   b. Initially activates the fast twitch muscle fibres.
   c. Activates both slow & fast twitch muscle fibres.
   d. Activates only the fast twitch muscle fibres.

5. Alteration in the conductivity of the nerves under the influence of constant direct current is called ....
   a. Electrophoresis.
   b. Accommodation.
   c. Phonophorosis.
   d. Electrotonus.

6. Pick the 'True' statement
   a. Electrically stimulated contractions are fatigue –resistance.
   b. Electrically stimulated contractions are more fatiguing hence longer rest periods should be provided.
   c. Slow twitch muscle fibers are frequently recruited during electrical stimulation.
   d. With electrical stimulation, nerve fiber with the smallest axonal diameter which innervates the slow twitch muscle fiber is stimulated.

7. The 'beat frequency' current is ……..
   a. High frequency current.
   b. Medium frequency current.
   c. Low frequency current.
   d. Rebox current.

8. In medical ionization, the usage of ______ ion is found to be effective in fungal infection.
   a. Iodide.
   b. Zinc.
   c. Salicylate.
   d. Copper.

9. Chronaxie of ………
   a. Innervated muscle is of very long.
   b. Denervated muscle is less than that of innervated.
   c. Denervated muscle is more than that of innervated.
   d. Both a and b.

10. ……. is the contraindication for cryotherapy.
   a. Cryglobulinemia.
   b. Raynaud's disease.
   c. Paroxysmalcold hemoglobinuria.
   d. All of the above.

CHAROTAR UNIVERSITY OF SCI. & TECHNOLOGY
CHARUSAT CAMPUS CHANGA

Examination Semester ________

Subject ________ No. ________

Candidate's Seat No. ________

Section ________

Date ________

Sr. Supervisor's Sign.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Physiotherapy (B.P.T)

Semester – V Examination – Nov 2011

PT301 ELECTROTHERAPY II

Date: 30.11.11, Wednesday **Time: 1.50p.m to4.30 p.m** **Maximum Marks: 60**

Instructions:

1. There are two parts in the paper. **PART –I** contains 10 Questions and **PART – II** contains 4 Questions.
2. **PART –I** should be answered in MCQ 'S question paper & **PART – II** should be answered in two separate answer books which has two sections.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **PART – II** is 2 Hr 40min.
5. Draw the figure wherever necessary.

PART-II

SECTION –I

Q.1. Long Question (1 out of 2) **(1× 10= 10)**

Describe determinants of a normal strength duration curve. Describe SD curve of Extensor digitorum taken at 12^{th} day, 3 months & 6 months in a patient with Radial nerve injury?

OR

Describe physiological effects of faradic currents & explain the applications of these effects in physiotherapy.

Q.2. Write short notes on: (4 out of 5) **(4× 5 = 20)**

1. Propogation of Action Potential.
2. Indications & Contraindications of Biofeedback.
3. Motor Unit.
4. Electric Shock – Safety & immediate treatment.
5. Electrical stimulation for denervated muscle.

SECTION –II

Q.3. Long Question (1 out of 2) **(1× 10= 10)**

Write in detail about the production of IFC. How will you treat the patient with hip joint osteoarthritis using IFT?

OR

Electrophysiological effects of IFC

Q.4. Write short notes on: (4 out of 5) **(4× 5 = 20)**

1. Pain gate theory.
2. Russian currents.
3. Therapeutic effects of cryotherapy.
4. Application principles in Iontophoresis.
5. Combination therapy.